

GOWTHAM P H

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SUMMARY

Full Stack AI Engineer with experience building LLM-integrated apps, RAG pipelines, and agentic workflows using Python, Java, and JavaScript. Certified in AWS, Azure AI, and Oracle GenAI. Skilled in AI automation with n8n — focused on shipping production-grade, AI-native systems.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript (ES6+), SQL
- **AI / LLM:** LangChain, Prompt Engineering, Structured Outputs, Tool-Calling, RAG Pipelines, Agentic Workflows, LLM Orchestration
- **Frontend / Backend:** React, Node.js, HTML5, CSS3
- **ML / Data:** Scikit-learn, TensorFlow, Keras, NLP, Pandas, CNN
- **Tools & Cloud:** Git, GitHub Copilot, Docker, n8n, Postman, Antigravity

PROJECTS

Java Student Management System (CRUD)

Jan 2025 – Feb 2025

Project Link

- Built a Java-based CRUD system using **Core Java** and **MySQL** for efficient student record management.
- Implemented **JDBC** and optimized SQL queries for insert, update, delete, and search operations.
- Designed modular code structure to improve maintainability and backend data handling.

Sign Language Recognition Using Deep Learning

Feb 2024 – Apr 2024

Project Link

- Built a CNN-based gesture recognition model using **TensorFlow** and **Keras**.
- Applied data augmentation and preprocessing to improve model generalization.
- Increased recognition accuracy by **15%** for accessibility-focused applications.

Agentic AI Workflow Automation — n8n & LangChain — GitHub

2025

Stack: Python, LangChain, n8n, OpenAI API, ChromaDB, FastAPI, Streamlit

- Designed a multi-step **agentic workflow** using **LangChain agents** + **n8n** to automate document processing and data extraction; implemented tool-calling for dynamic REST API invocation and **ChromaDB** vector queries — cut manual execution time by ~70%
- Delivered a **Streamlit** dashboard for real-time agent monitoring, trace visualisation, and cost/latency tracking per workflow run

EXPERIENCE

Machine Learning Intern — Coding Raja Technologies

Jun 2024 – Aug 2024

- Trained Random Forest, XGBoost, and K-Means models on a 50K-row dataset; improved F1-score from 0.74 to **0.87** via GridSearchCV hyperparameter tuning
- Applied SMOTE + PCA to resolve class imbalance, boosting minority-class recall by **22%** on a fraud detection task; delivered reproducible ML pipelines with Pandas and Scikit-learn

EDUCATION

B.Tech in Artificial Intelligence & Data Science — Panimalar Engineering College

2022 – 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Machine Learning, Artificial Intelligence

CERTIFICATIONS

Oracle Cloud Infrastructure Generative AI Certified Professional | Microsoft Azure AI Fundamentals (AI-900) | AWS Certified Cloud Practitioner | Full Stack Web Development (React, Node.js, REST APIs) | Prompt Engineering for LLM Applications

ACHIEVEMENTS & ACTIVITIES

- Built 3+ AI-powered full stack applications (RAG, agentic workflows, LLM apps) deployed with real-world use cases
- Solved 80+ DSA problems across platforms like LeetCode and GeeksforGeeks; strong in problem-solving and optimization
- Actively explored AI tools (LangChain, n8n, OpenAI APIs) and implemented production-level features like tool-calling and structured outputs